

Original Article

Low Vasopressin in Myalgic Encephalomyelitis/Chronic Fatigue Syndrome



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ABSTRACT

Objective: The complex clinical picture of myalgic encephalomyelitis (ME)/chronic fatigue syndrome (CFS) includes orthostatic intolerance with symptoms and signs suggesting abnormal water homeostasis and hypovolemia. Since many ME/CFS patients report polyuria-polydipsia, we conducted an observational study based on series of plasma and urine osmolality (P-Osm and U-Osm) as well as plasma levels of vasopressin (P-VP) or copeptin in consecutive patients diagnosed with ME/CFS according to the International Consensus Criteria.

Methods: P-VP as well as P-Osm and U-Osm were measured in 111 patients after 10 h overnight fasting and fluid deprivation. Additional 13 patients were assessed for copeptin, when P-VP measurements were no longer available. The clinical routine also included brain MRI and blood chemistry.

Results: P-Osm was abnormally high (>292 mOsm/kg) in 71 of 124 (57.3%) and U-Osm below the reference interval (<750 mOsm/kg) in 82 of 124 (66.1%) patients. P-VP was below the level of detection (<1.6 pg/mL) in 91 of 111 (82.0%) patients. A normal P-VP level compared with their P-Osm was found in 11 of 111 (9.9%) patients. Copeptin levels were all within the given reference range, albeit in the lower end in most patients. No indication of relevant pathology in either hypothalamus or hypophysis was present.

Conclusions: Our findings suggest that chronic down-regulation of VP mimicking central diabetes insipidus is an important measurable part of the disease mechanism that potentially contributes to criterial symptoms of ME/CFS.

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Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is an illness characterized by intolerance to upright posture and exertion. Even low intensity physical and/or cognitive exertion triggers postmalaise, long-lasting or persistent impairment of symptoms including pain, sleep disturbance, neurological, autonomic, gastrointestinal and/or immunologic symptoms (Table 1).

Abbreviations: cDI, central diabetes insipidus; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; P-Osm, plasma osmolality; (P-)VP, (plasma) vasopressin; U-Osm, urine osmolality.

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ME/CFS is not a rare condition, with prevalence estimation of 0.89% of the population.¹ Still, its pathophysiology remains elusive.

Vasopressin (VP), also called antidiuretic hormone, has a central role in the regulation of water balance and maintenance of both sufficient fluid volume and blood pressure, which are prerequisites of orthostatic tolerance.² VP synthesized by neurons in the hypothalamus is secreted from the posterior lobe of the hypophysis as a response to increased plasma osmolality. The hypothalamic osmoreceptors are very sensitive: only 1% rise in plasma osmolality triggers VP secretion, and VP exerts an antidiuretic effect in the kidneys via V2-receptors leading to water retention and normalizing plasma osmolality.^{3,4} Through V1 receptors expressed in blood vessels, VP facilitates vasoconstriction.⁵ Finally, VP acts in concert with adrenergic neural activity in locus coeruleus, with potential implications for ME/CFS symptomatology, as central noradrenergic activity rises when upright, maintaining a tonic stimulation of vasopressin secretion.^{6–8}

Hypovolemia is described among patients with ME/CFS.^{9–11} Patients with ME/CFS often describe excessive thirst and high urine volumes, symptoms that are regularly dismissed if not specifically asked about. We report a clinical observational study on a series of osmolality and VP/copeptin levels in 124 patients with ME/CFS and discuss the potential role of disturbed VP function in ME/CFS symptomatology.

Materials and Methods

This retrospective cross-sectional study of consecutive patients with ME/CFS was carried out at The Bragée ME-center in Stockholm, Sweden. Between March 2019 and November 2020, our neurologist (H.H.) met 173 consecutive patients with ME/CFS diagnosed according to the International Consensus Criteria.¹² Thirty-two patients were ineligible due to i) diabetes mellitus ($n = 5$); ii) severe rheumatoid arthritis ($n = 1$); iii) coming from outside the Stockholm region ($n = 6$); iv) inability to undertake overnight water deprivation and/or holding urine ($n = 10$); v) inability to reach a laboratory within the time-frame ($n = 6$); or vi) miscellaneous reasons ($n = 4$).

As standard clinical routine, all 141 recruitable patients delivered morning samples of plasma VP (P-VP), plasma osmolality (P-Osm) and urine osmolality (U-Osm) after 10-hour water deprivation and fasting. In 12 patients, the results were incomplete due to technical problems or the patient's inability to hold urine overnight. In the last 13 patients, copeptin measurement was delivered instead of VP, as RIA measurements were in late 2020 withdrawn to minimize the use of radioactive reactants, and copeptin mirrors the level of VP.¹³ Thus, out of the laboratory requests for 141 patients, reliable measurements were completed in 124 patients (111 with P-VP, and 13 with P-copeptin).

All laboratory assessments were done by an accredited laboratory (osmolalities and VP at Karolinska University Hospital, copeptin at Linköping University Hospital). Assessment of osmolality was done according to classical freeze/thawing procedures. VP was analyzed by a radioimmunoassay with an intra-assay coefficient of variation less than 4% and an ability to detect vasopressin at the level of 1.6 pg/ml. Copeptin was measured immunochemically with a ThermoFisher (BRAHMS Kryptor Gold) instrument that bases on the TRACE-technics (Time-Resolved Amplified Cryptate Emission). The reported measurement error is about 6% to 9% with low levels (around 5 pmol/L) and about 5% at high levels (95 pmol/L).

The standard diagnostic procedure included brain MRI and measurements of plasma glucose, sodium, and potassium, as well as serum thyroid stimulating hormone and cortisol.

The demographic data of the 124 study participants are shown in Table 2. Sex distribution, age ranges and body mass indices were typical of patients with ME/CFS. The average/median disease duration was 7.3 or 7.5 years. None of the participants were taking desmopressin or lithium.

Ethical approval was granted by the Swedish Ethical Review Authority (nr. 2021-03114).

Results

Of all 124 patients, P-Osm was high (>292 mOsm/kg) in 71 (57.3%) and U-Osm low (<750 mOsm/kg) in 82 (66.1%) patients.

The combined results of P-VP, P-Osm, and U-Osm of the 111 VP-patients are presented in Fig. 1. P-VP levels remained very low in most patients, with levels under detection threshold (1.6 pg/ml) in 86 out of 111 (77.5%). Urine osmolality was low (<750 mOsm/kg) in 74 out of 111 (66.7%). Eleven patients (9.9%) had a normal level of

Highlights

- Orthostatic intolerance and polyuria-polydipsia typify disabling forms of myalgic encephalomyelitis (ME)/chronic fatigue syndrome (CFS)
- In a consecutive series of ME-CFS patients, majority had very low vasopressin levels, relatively high plasma osmolality and low urine osmolality, in absence of overt hypothalamic or hypophyseal pathology
- Chronic down-regulation of vasopressin mimicking central diabetes insipidus may contribute to the core symptoms of ME/CFS

Clinical Relevance

Myalgic encephalomyelitis/chronic fatigue syndrome is a prevalent, chronic and highly debilitating syndrome often affecting working-age individuals, adolescents, and even children. There are no reliable biological markers for the disease and no proven evidence-based therapies. Further elucidation of the pathophysiological role of vasopressin could help clarify its diagnostic and therapeutic value.

P-VP in relation to their P-Osm, and 5 of them (4.5%) were within the reference range for all the 3 parameters. Finally, 57 individuals had elevated P-Osm with simultaneous pathologically low levels of P-VP, ie lower-right area of 'central diabetes insipidus' in Fig. 1.

Copeptin levels were all within the given reference range, albeit in the lower end in most patients (Figs. 2 and 3).

Blood chemistry including sodium, potassium, glucose, TSH and cortisol levels were within normal range. MRI scanning was performed in 111 patients with VP measurements. In 2 additional cases, MRI was either not obtained or not available. Among the completed scans, 3 showed an empty sella, while the remaining studies were considered normal. In the group of 13 patients with copeptin measurements, one had a 2-mm pituitary (adenohypophyseal) cyst, one scan could not be obtained, and the rest were normal. Accordingly, the frequency of abnormal hypophyseal MR findings among the scans was 3.6%; none showed laboratory findings consistent with panhypopituitarism.

There was no correlation between P- and U-Osm (Fig. 4) depicting the failure of rising P-Osm to stimulate excretion of VP in this patient cohort. Fourteen individuals had both P-Osm and U-Osm within the reference range (Fig. 4, shaded area).

The mean copeptin level in all subjects was 7.6 pmol/l (SEM 1.49). More detailed results in both genders are reported in Table 2. Copeptin over plasma osmolality was within reference range in all the 13 patients, with a positive correlation with U-Osm ($R^2 = 0.619$, Fig. 4). Copeptin 7.5 pmol/l, which is within the reference range for copeptin, turned out to be a cut-off level distinguishing those patients with abnormally low U-Osm from those with normal U-Osm.

Discussion

Our results reveal a high prevalence of low P-VP, low urine osmolality and high plasma osmolality after overnight water restriction. These findings are in line with polydipsia-polyuria, which is a common complaint when specifically asked about during the clinical interview. Polydipsia-polyuria often appears at the beginning of illness, and the manifestations may be constant and monophasic, transient, or intermittent, and it seems to be

Table 1
The Cardinal Criteria for ME-CFS (According to International Consensus Criteria)

Category	Requirement	Key features
1. Postexertional neuroimmune exhaustion	mandatory	a marked, rapid physical/cognitive fatigability b postexertional exacerbation c prolonged recovery (≥ 24 h) d low threshold of physical/mental fatigability (a significant reduction in pre-illness activity level)
2. Neurological impairments	(≥ 1 symptom from ≥ 3 subcategories)	a cognitive impairments (difficulty processing information, impaired memory) b pain (headache, myalgia, noninflammatory pain) c sleep disturbance (unrefreshing sleep, circadian dysregulation) d motor/sensory issues (perceptual, motor)
3. Immune, gastrointestinal & genitourinary impairments	(≥ 1 symptom from ≥ 3 subcategories)	a flu-like symptoms (recurrent sore throat, feverishness) b susceptibility to viral infections c GI symptoms (nausea, IBS-like, bloating, abdominal pain) d GU symptoms (frequency/urgency, nocturia) e sensitivities (food/chemical/odours)
4. Energy production/transport impairments	(≥ 1 symptom from ≥ 1 subcategories)	a cardiovascular (orthostatic intolerance, POTS, palpitations) b respiratory (air hunger, dyspnea, fatigue) c loss of thermostatic stability (sweating, temperature instability, cold extremities) d intolerance to extremes of heat/cold

Abbreviations: GI = gastrointestinal; GU = genitourinary; IBS = irritable bowel syndrome; POTS = postural orthostatic tachycardia syndrome. According to additional criteria, the duration of symptoms must be at least 6 mo, and other medical or psychiatric conditions must not explain the symptoms.

Table 2
Demographic Data and Blood Chemistry of the Patient Population

	Women (n = 105)			Men (n = 19)		
	n	Mean	Range	n	Mean	Range
Age (y)	105	43.3	14-66	19	42.7	16-65
BMI (kg/m ²)	97	25.0	17.1-36.9	18	27.2	19.6-35.5
Disease duration (y)	105	7.2	0.5-11	19	8.2	0.5-11
P-Na (mmol/l)	90	140.1	134-144	19	141.3	137-144
P-K (mmol/l)	94	4.0	3.4-4.6	19	4.0	3.7-4.8
P-glucose (mmol/l)	102	5.5	4.2-8.6	19	5.6	4.7-6.3
S-TSH (mU/l)	95	2.3	0.1-9.6	19	2.4	1.0-7.5
S-cortisol (nmol/l)	100	417	161-1058	18	409	195-712
Osmolality (mOsm/kg)						
Plasma	105	293	279-318	18	295	286-310
Urine	105	606	144-1084	19	730	290-1197
P-vasopressin						
Below detection (n)	76	72%		10	53%	
Measurable (pg/ml)	19	2.57	1.5-7.6	6	2.32	1.7-4.2
P-copeptin (pmol/l)	10	6.6	2.7-15	3	11.5	3.4-18

Abbreviations: BMI = body mass index; TSH = thyroid stimulating hormone.

often overlooked in the patient-physician communication. The combination of high P-Osm with low P-VP excludes both psychogenic polydipsia and renal resistance to VP. Instead, it indicates a central VP deficiency resembling central diabetes insipidus (cDI), a rare condition with an estimated prevalence of 1:25 000, which is less than one percent of the estimated prevalence of ME/CFS.¹⁴ In general, cDI is mainly the result of destruction of the producing hypothalamic or secreting hypophyseal cells. In absence of structural abnormalities causing such a condition, cDI is considered idiopathic. The high incidence of central-type of VP deficiency in our patient cohort suggests that diminishing of the tonic stimulation of VP secretion, which in a healthy person is maintained by central noradrenergic activity, may diminish the sensitivity of the osmoreceptors and cause low VP secretion in ME/CFS.

Low VP in chronic fatigue is not a novel finding. As early as in 1993 Bakheit et al reported a low VP despite high P-Osm in 9 patients with postviral fatigue.¹⁵ Wyller et al had similar findings in 65 adolescents with ME/CFS.¹⁶ A small controlled study using essentially the same diagnostic criteria but with a considerably younger patient population and a treatment arm found not only significantly lower levels of antidiuretic hormone (VP) and renin

activity but also consistently lower cardiac parameters and stroke volume index.¹⁷ Our study of 124 consecutive ME/CFS patients corroborates the observation of significantly lower VP levels of these previous reports, which may not have been given due consideration over the years.

VP is of special interest in ME/CFS due to its contribution to orthostatic tolerance by several parallel mechanisms. Firstly, VP is essential for sufficient blood circulation as it maintains blood volume by stimulating water reabsorption in renal distal tubules, and it creates peripheral resistance by constricting peripheral blood vessels. A moderate decrease in blood volume (5% to 10%) causes a significant increase in VP levels primarily through baroreceptor reflex.¹⁸ Thus, VP deficiency gives rise to simultaneous absolute hypovolemia and pooling of blood to lower extremities in upright position. This pathology predisposes any patient to a compensatory postural tachycardia syndrome, postural orthostatic tachycardia syndrome, one of the few objective findings in ME/CFS.^{19,20} Secondly, VP also acts as a neurotransmitter in close interaction with locus coeruleus and the adrenergic neural network of dorsal pontine area, which are of great importance for maintaining vigilance and muscle tone. Indeed, according to Pompeiano⁶'s classical experiments, VP has a role in controlling posture and the gain of vestibulospinal reflex. Thus, in addition to the circulatory adjustments, the neurotransmitter VP supports vigilance and limb extensor muscular tone opposing the gravitational pull, reinforcing orthostatic tolerance in the awake state.

VP is also involved in several other processes relevant in ME/CFS. As a hormone, it enhances the release of adrenocorticotrophic hormone in both instances secretion.^{21,22} Unexplained hypothalamic-pituitary-adrenal axis hypoactivity has been described in patients with ME/CFS, and acute adrenocorticotrophic hormone in both instances receptor agonism has according to Pereira et al resulted in sustained symptom improvement in ME/CFS.²³⁻²⁵ Furthermore, VP is engaged in maintaining the diurnal rhythm.²⁶ Indeed, sleep dysfunction, including unrefreshing sleep, inadequate sleep, or sleep rhythm disturbances, is obligatory in diagnosing ME/CFS according to the International criteria. Furthermore, VP is also involved in energy metabolism as peripheral enzymatic degradation of VP is linked with glucose intake in adipose and muscle cells.²⁷ Consequently, low VP levels may have a bearing on energy metabolism of musculature.²⁸

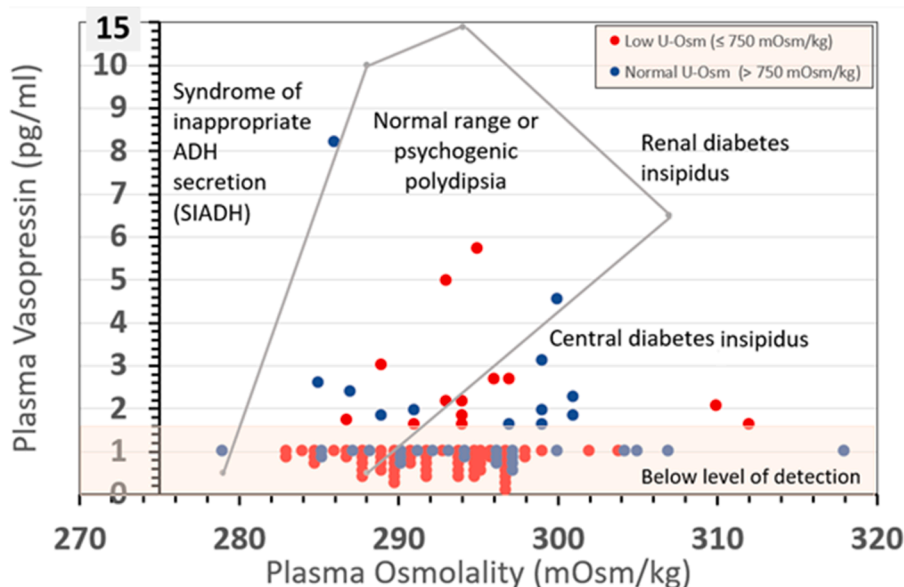


Fig. 1. The reference interval of P-VP is semi-linearly correlated with the simultaneous level of P-Osm, ie normal pituitary and renal function (adapted from Tietze Textbook of Clinical Chemistry and Molecular Diagnostics 2006, fourth Edition, p. 1994). Variable combinations of P-VP and P-Osm outside the pentagon are diagnostic for distinct subtypes of VP disorders. The syndrome of inappropriate VP secretion (SIADH) is characterized by lacking inhibition of VP secretion by hypo-osmolality. In renal diabetes insipidus (rDI), kidneys do not respond to VP, and the patient becomes hyperosmolar despite high VP values. In central diabetes insipidus (cDI), the combination of hyperosmolality and low VP reflects a condition where high plasma osmolality fails to stimulate VP secretion. Each dot represents a study patient. All blue dots are patients with normally concentrated urine (U-Osm > 750 mOsm/kg). The blue dots within the pentagon with measurable levels of P-VP represent patients with all 3 parameters within references (which represent the control values of strictly healthy reference range). The red dots indicate patients with low urine osmolality (≤ 750 mOsm/kg). Most of the patients have very low P-VP in line with low U-Osm, despite pathologically high P-Osm. Osm = osmolality; P-VP = plasma levels of vasopressin.

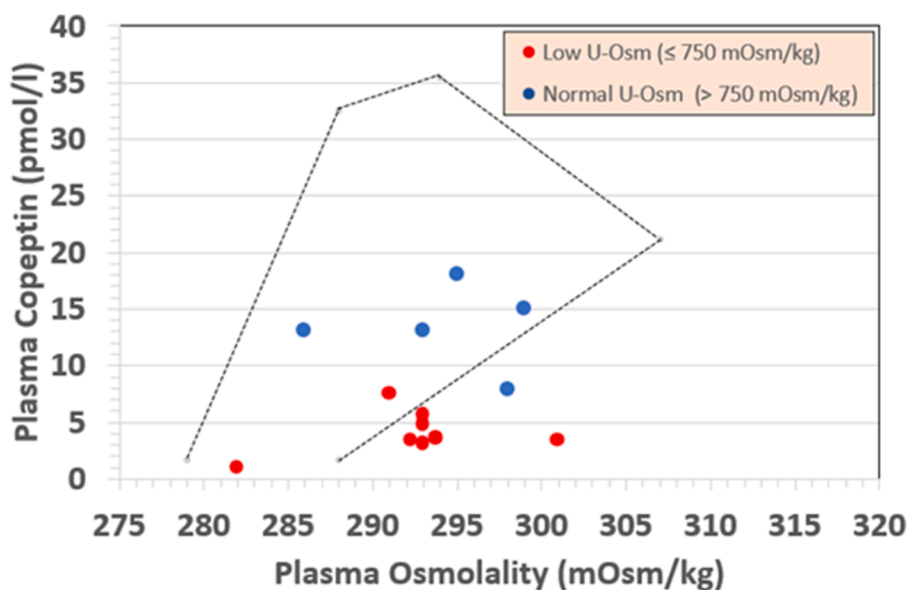


Fig. 2. The relationship between plasma osmolality and copeptin in 13 patients. The 4 blue dots within the pentagon represent the patients with all the 3 measures (P-copeptin, P- and U-Osm) within reference. The red dots are those with low urine osmolality (≤ 750 mOsm/kg). In line with the pattern of P-VP, most of the red dots have low copeptin despite high P-Osm after overnight water deprivation. The pentagon reference area (dotted line) is hypothetical, mathematically reconstructed from the data with the assumption that copeptin and VP have a similar normal relationship with plasma osmolality (shown in Fig. 1). VP = vasopressin; Osm = osmolality; P-VP = plasma levels of vasopressin.

Chronic hyperosmolality ensuing from recurrent or sustained down-regulation of VP secretion has adverse effects on several functions. Hyperosmolality attenuates post-exercise sweating, providing a reasonable explanation for the elevated body temperature or sensation of fever typically reported by ME/CFS patients after exertion.²⁹ Elevated tissue temperature combined with hypovolemia is detrimental to aerobic performance.³⁰ On cellular

level, the compensatory mechanisms may be altered by hyperosmolar extracellular fluid: Esfandarypour and colleagues have shown that exposing peripheral blood mononuclear cells of ME/CFS patients to hyperosmotic stress reveals an impedance modulation response profoundly different from that of healthy controls.³¹ Furthermore, hyperosmolality may even contribute to sensitization of neurons, an essential feature in ME/CFS

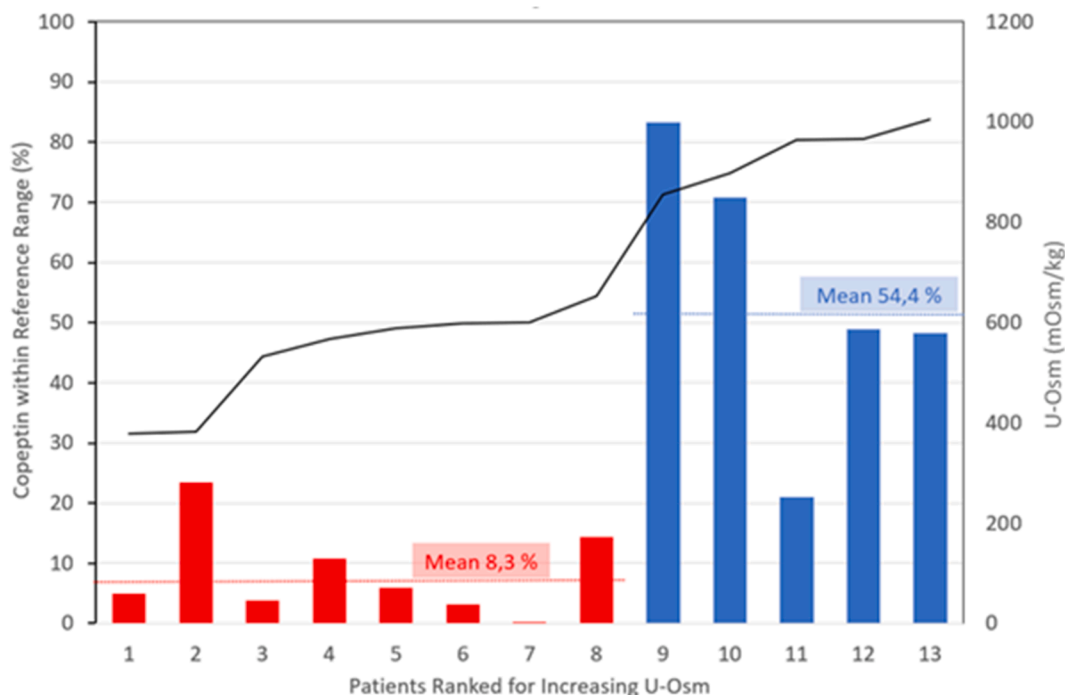


Fig. 3. Copeptin in patients ranked according to their U-Osm after fluid deprivation (from low till normal, shown by the black line with representative y-axis on the right). Copeptin in all patients studied were within their reference ranges for P-Osm. However, copeptin levels in patients with low U-Osm (≤ 750 mOsm/kg) were just marginally (8.3%) above the lower reference limit (red bars), in the patients with normal U-Osm (>750 mOsm/kg, blue bars), they were on average in the middle of their corresponding reference ranges (54.4%). In the presence of normal VP/copeptin, the kidneys of ME/CFS patients respond normally by concentrating urine, speaking against VP resistance as the mechanism of low U-Osm. Osm = osmolality; ME/CFS = myalgic encephalomyelitis/chronic fatigue syndrome; VP = vasopressin.

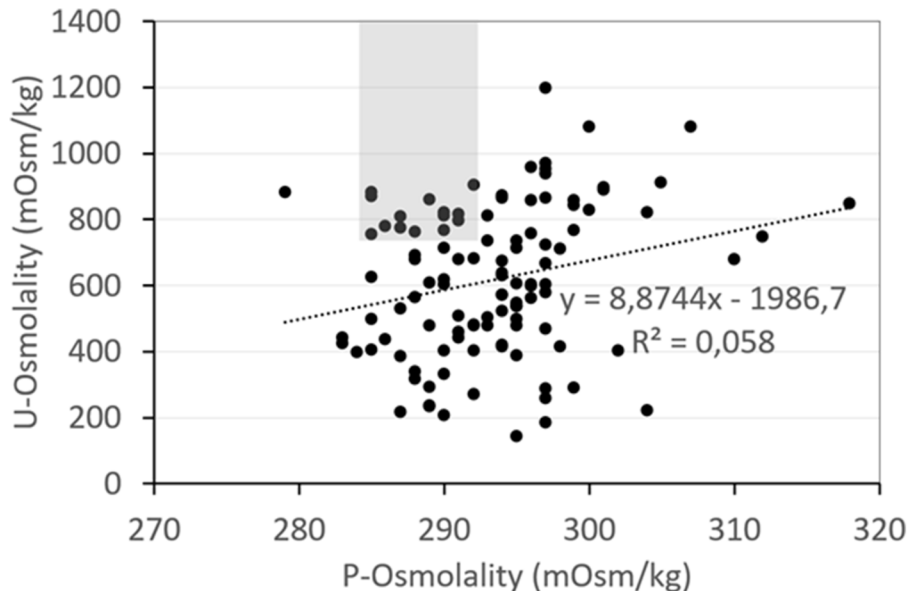


Fig. 4. Relationship between U-Osm and P-Osm in 111 patients, with ME/CFS following 10-hour fluid deprivation. Shaded area represents the reference ranges for P-Osm (285–292 mOsm/kg) and U-Osm (>750 mOsm/kg). Osm = osmolality; ME/CFS = myalgic encephalomyelitis/chronic fatigue syndrome.

pathophysiology, since influx of cations into shrinking neurons might shift the resting potential closer to the threshold of action potential.^{32,33} The resulting excessive irritability of different types of neurons could explain a wide range of neurological symptoms such as spontaneous pain and oversensitivity to light and sound. Finally, the role of VP in ME/CFS could explain why some patients

with ME/CFS and/or postural orthostatic tachycardia syndrome benefit from recurrent intravenous saline infusions, ie treatments targeted against both hypovolemia and hyperosmolality.³⁴ Furthermore, an important treatment implication would be the potential of vasopressin analogues. As suggested by a previous study, part of the patient population may benefit

from the longer-acting desmopressin.¹⁷ However, desmopressin mostly lacks the effect on blood vessel constriction, and its clinical application warrants further study in this group of patients.

The limitation of the study is the cross-sectional, observational nature of the series as an aside of clinical work, precluding the use of recruited healthy controls. An additional limitation is the extent of missing data; 10 patients who were able to visit the ME-center were unable to even undertake the regimen with overnight fluid deprivation and holding of urine, and further 17 patients were not able to fully comply with it. In addition, we obviously could not recruit the most severely affected, completely bedbound patients not tolerating upright position at all. On the other hand, this illustrates the debilitating nature of both polydipsia-polyuria and orthostatic intolerance in ME/CFS, and the dropouts may be expected to represent the severe end of the spectrum. A pragmatic limitation in the future may be the availability of VP measurements. Copeptin is secreted in equimolar amount to VP, and whilst VP has a short half-life (5–20 min), copeptin has shown to be a more stable molecule reflecting long-term VP levels.³⁵ Thus, it might not be as sensitive to abnormal dynamic changes of VP secretion. Due to the small size of our copeptin subpopulation, it is not possible to conclude whether copeptin measurements are equally informative in the diagnostics of this condition.

The strengths are the large consecutive cohort of carefully diagnosed ME/CFS patients, as well as the relative consistency of the findings. A further strength is that we were able to measure vasopressin levels directly in most patients, rather than relying on copeptin.

Our findings underscore the need for careful history taking in clinical practice, where polydipsia-polyuria provide further support for the diagnosis after exclusion of diabetes mellitus and related differential diagnoses. Further study should be aimed at determining the diagnostic value of low U-osm after 10-hour overnight water deprivation, if necessary in combination with copeptin/VP/P-Osm, in screening or even as a biomarker for a ME/CFS subpopulation.

The high incidence of low VP in our patient cohort suggests that chronic down-regulation of VP secretion may be an elementary part of pathophysiology of ME/CFS, first and foremost the chronic hypovolemia which hitherto has remained unexplained. Finally, understanding the underlying pathophysiology will be essential, including the role of autoimmunity as suggested by a previous case-control study on antipituitary and antihypothalamic antibodies.³⁶ Given the multitude of the functions of VP as a hormone and a neurotransmitter, a chronic VP deficiency could be the reason for a wide array of the criteria symptoms of ME/CFS. Ultimately, a deeper understanding of this pathophysiology has the potential to inform more effective therapeutic strategies for the patients affected.

Disclosure

The authors have no conflicts of interest to disclose.

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